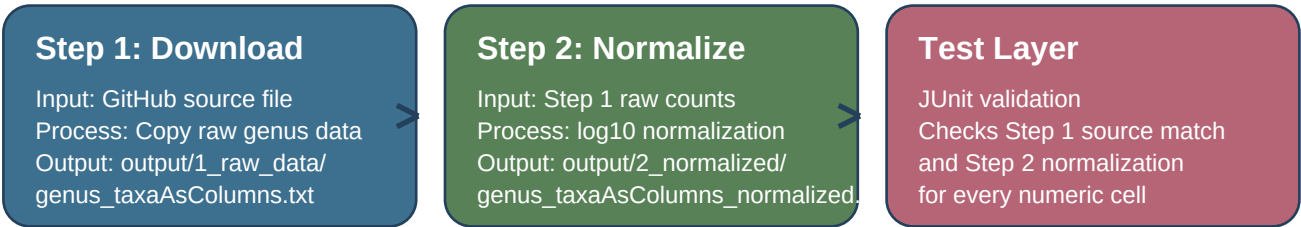


OrchestraSimpleDemo Test Report

Generated on April 12, 2026

Current Orchestration Pattern



The pipeline currently orchestrates a raw-data acquisition step followed by normalization, with JUnit tests validating both source fidelity and normalized-value correctness.

Test Summary

This report summarizes the Java/JUnit testing completed so far for the OrchestraSimpleDemo bioinformatics pipeline. The current automated checks cover both raw-data integrity and normalized-output correctness.

Test ID	Scope	Result	Evidence
T1	Step 1 raw-data validation	PASS	The file output/1_raw_data/genus_taxaAsColumns.txt exactly matched the
T2	Step 2 normalization validation	PASS	Every numeric cell in output/2_normalized/genus_taxaAsColumns_normalized

Execution Notes

The JUnit suite was executed successfully on April 12, 2026. One test downloads the canonical source file from GitHub at runtime and compares the remote bytes with the local Step 1 output bytes. A second test recomputes normalized values from the raw count table and verifies every numeric cell in the Step 2 output.

Reference source:
https://raw.githubusercontent.com/kwinglee/UrbanRuralChina/master/16SrRNA/inputData/RDP/genus_taxaAsColumns.txt

Current Coverage

Coverage now includes both current orchestration steps: verification that the downloaded raw data file is an exact mirror of the expected source, and verification that normalized output values satisfy the stated formula across the entire table. Future tests can extend this to end-to-end pipeline execution and additional downstream stages.